

Auctioning (EU ETS 1)

German Auctioning of Emission Allowances

Fourth Quarter 2024

Umwelt Bundesamt

DEHSt
Deutsche Emissionshandelsstelle

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German Auctioning of Emission Allowances

The German auctions of the fourth trading period of the European Emissions Trading System 1 (EU ETS 1) are held weekly on the European Energy Exchange (EEX) spot market. The German Emissions Trading Authority (DEHSt) at the German Environment Agency is the competent authority for the German auctions. Further updated information on the fourth trading period auctions can be downloaded from the EEX website. Reports on the second, third and fourth trading periods auction results are available on the DEHSt auction website. The 2024 and 2025 auction calendars are published on the EEX website.

The following tables summarise the key data of the German auctions on the EEX in 2024 and 2025:

2024		
Auction product	Fourth trading period **EUA spot contracts**	Fourth trading period **EUAA spot contracts**
Period	19/01/2024 to 13/12/2024	16/10/2024
Auction volume entire year	83,991,500 EUA[1]	1,061,000 EUAA
Auction volume per auction	January–August[2]: 1,796,000 EUA **September–December: 1,886,500 EUA**	1,061,000 EUAA on 16/10/2024
Auction frequency	Weekly (Fridays, 09:00 to 11:00 am)	On 16/10/2024 (09:00 to 11:00 am)
Bid volume	500 EUA/EUAA	
Auction mode	Single-round, uniform-price procedure with a closed order book	

2025	
Auction product	Fourth trading period **EUA spot contracts**[3]
Period	10/01/2025 to 12/12/2025
Auction volume entire year	96,764,500 EUA (provisional)[4]
Auction volume per auction	January–August: 1,607,000 EUA September–December: 3,129,500 EUA (provisional)[4]
Auction frequency	Weekly (Fridays, 09:00 to 11:00 am)
Bid volume	500 EUA
Auction mode	Single-round, uniform-price procedure with a closed order book

1 The MSR reduction for the auction volumes from September 2024 to August 2025 was published by the European Commission on 01 June 2024.

2 From 2024 onwards, the volumes to be auctioned in August will no longer be reduced by half as in the past. This is in line with the requirements of the new EU Auctioning Regulation.

3 From 2025 onwards, aviation allowances (EUAA) will no longer be auctioned on the primary market. The corresponding volumes will be auctioned proportionally as EUA spot contracts in the weekly auctions. This is in line with the requirements of the new EU Auctioning Regulation.

4 The MSR reduction for the auction volumes from September 2025 to August 2026 is expected to be determined by the European Commission in June 2025.

1 Overview

1.1 Fourth quarter of 2024

Between October and December, eleven auctions of emissions allowances (EUA) of the fourth trading period were successfully carried out for Germany at the EEX.

Overall, about **21 million EUA** with a total value of almost **€ 1.36 billion** were auctioned. This results in a volume-weighted average price per allowance of **€ 65,52** (minus 3 percent compared to the previous quarter). The deviations of the clearing prices from the current prices on the secondary market fluctuated between −0.36 and +0.01 percent (see also Chapter 3.3).

Overall, valid bids totalling about 40 million allowances were entered into the trading system across the eleven EUA auction dates. Thus, the average cover ratio in the reporting period was 1.95 (see also Chapter 2.2). The average participation number of 23 bidders was at the level of the previous quarter (23). The average number of successful bidders was above the previous quarter's level (17) at 18, (see also Chapter 2.1).

The following table gives an overview of the most important auction figures from the EUA auctions performed during the reporting period.

Table 1: Overview of German EUA auctions in the fourth quarter of 2024

Date	Auction volume	Bid volume	Cover ratio	Bidders	Successful bidders	Price	Revenue
04/10/2024	1,886,500	3,497,500	1.85	19	17	€ 61.03	€ 115,133,095
11/10/2024	1,886,500	3,826,500	2.03	25	17	€ 63.71	€ 120,188,915
18/10/2024	1,886,500	3,810,500	2.02	24	17	€ 61.90	€ 116,774,350
25/10/2024	1,886,500	3,758,500	1.99	21	18	€ 66.32	€ 125,112,680
October	**7,546,000**	**14,893,000**	***1.97**	***22**	***17**	***€ 63.24**	**€ 477,209,040**
01/11/2024	1,886,500	3,765,500	2.00	20	14	€ 64.46	€ 121,603,790
08/11/2024	1,886,500	4,022,000	2.13	22	17	€ 66.20	€ 124,886,300
15/11/2024	1,886,500	3,121,500	1.65	21	19	€ 66.78	€ 125,980,470
22/11/2024	1,886,500	4,056,500	2.15	23	20	€ 69.71	€ 131,507,915
29/11/2024	1,886,500	3,376,500	1.79	26	23	€ 67.30	€ 126,961,450
November	**9,432,500**	**18,342,000**	***1.94**	***22**	***19**	***€ 66.89**	**€ 630,939,925**
06/12/2024	1,886,500	3,757,000	1.99	25	19	€ 67.25	€ 126,867,125
13/12/2024	1,890,500	3,437,500	1.82	23	20	€ 66.07	€ 124,905,335
December	**3,777,000**	**7,194,500**	****1.91**	**24**	**20**	****€ 66.66**	**€ 251,772,460**
Q4 2024	**20,755,500**	**40,429,500**	****1.95**	***23**	***18**	****€ 65.52**	**€ 1,359,921,425**

Source: EEX, DEHSt
*average
** volume-weighted average

In addition, 1,061,000 aviation allowances (EUAA) with a total value of about € 68 million were auctioned for Germany on 16th October. The auction price for the 6 successful bidders was € 64.14. Overall, 15 bidders with valid bids totalling about 3.2 million allowances participated in the auction. The auction on 16th October was the last EUAA auction for Germany. From 2025 onwards, EUAA will no longer be offered on the primary market. In future, the corresponding volumes will be auctioned on a proportional basis as EUA spot contracts at the weekly auctions. This is in line with the requirements of the new EU Auctioning Regulation.

1.2 Overview of the entire year 2024

In 2024 a total of about **85 million EUA and EUAA** with a total value of about **€ 5.53 billion** were auctioned at the EEX for Germany. This results in a volume-weighted average price per allowance of **€ 65**.

The following table summarises the auction results on a monthly basis.

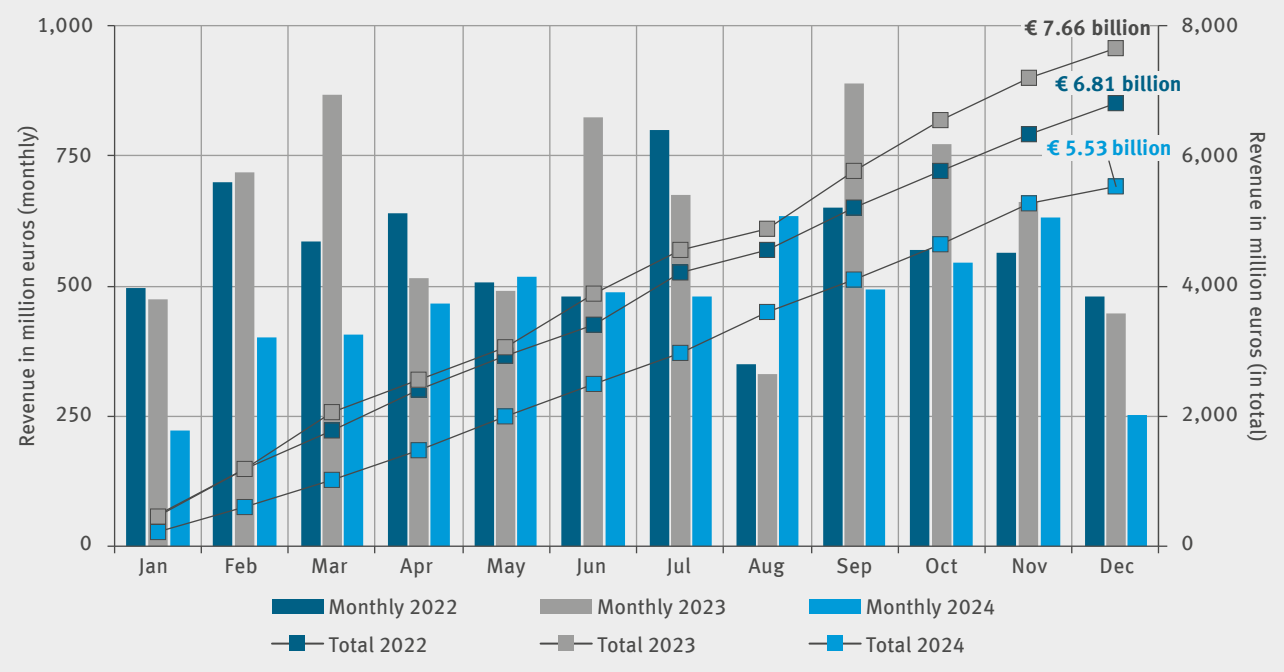
Table 2: Overview of the entire year 2024

Month	Type	Auction volume	Bid volume	Cover ratio	Bidders	Successful bidders	Price	Revenue
January	EUA	3,592,000	7,586,000	*2.11	*26	*9	*€ 61.94	€ 222,488,480
February	EUA	7,184,000	12,874,000	*1.79	*23	*21	*€ 55.70	€ 400,112,880
March	EUA	7,184,000	14,021,000	*1.95	*24	*17	*€ 56.60	€ 406,578,480
April	EUA	7,184,000	14,477,000	*2.02	*27	*17	*€ 64.65	€ 464,463,560
May	EUA	7,184,000	14,492,000	*2.02	*24	*14	*€ 71.98	€ 517,122,280
June	EUA	7,184,000	13,068,000	*1.82	*23	*18	*€ 67.67	€ 486,141,280
July	EUA	7,184,000	13,844,000	*1.93	*24	*18	*€ 66.70	€ 479,172,800
August	EUA	8,994,000	18,068,500	**2.01	*22	*16	**€ 70.40	€ 633,208,900
September	EUA	7,546,000	14,641,000	*1.94	*24	*16	*€ 65.13	€ 491,470,980
October	EUA	7,546,000	14,893,000	*1.97	*22	*17	*€ 63.24	€ 477,209,040
	EUAA	*1,061,000*	*3,212,000*	*3.03*	*15*	*6*	*€ 64.14*	*€ 68,052,540*
November	EUA	9,432,500	18,342,000	*1.94	*22	*19	*€ 66.89	€ 630,939,925
December	EUA	3,777,000	7,194,500	**1.91	24	20	**€ 66.66	€ 251,772,460
EUA		**83,991,500**	**163,501,000**	****1.95**	***24**	***17**	****€ 65.01**	**€ 5,460,681,065**
EUAA		***1,061,000***	***3,212,000***	***3.03***	***15***	***6***	***€ 64.14***	***€ 68,052,540***
Total		**85,052,500**	**166,713,000**	****1.96**	***23**	***17**	****€ 65.00**	**€ 5,528,733,605**

Source: EEX, DEHSt
* average
** volume-weighted average

The following figure shows the German EUA/EUAA auction revenues in the period since 2022. In addition to the monthly values, the cumulative total revenues of the individual years are also shown in the chart. German revenues in 2024 were about 28 percent below the record result from the previous year (2023: € 7.66 billion). This is mainly due to a significant drop in carbon market prices as well as lower auction volumes. From 2024, the volumes to be auctioned in August are no longer reduced by half as before in accordance with European regulatory requirements. This effect is clearly shown in Figure 1. Since 2012 German auction revenues flow entirely into the Climate and Transformation Fund (KTF)[5].

5 Until July 2022 – Energy and Climate Fund (EKF).



Source: EEX, DEHSt

Figure 1: Auction revenues (EUA+EUAA) from 2022 to 2024

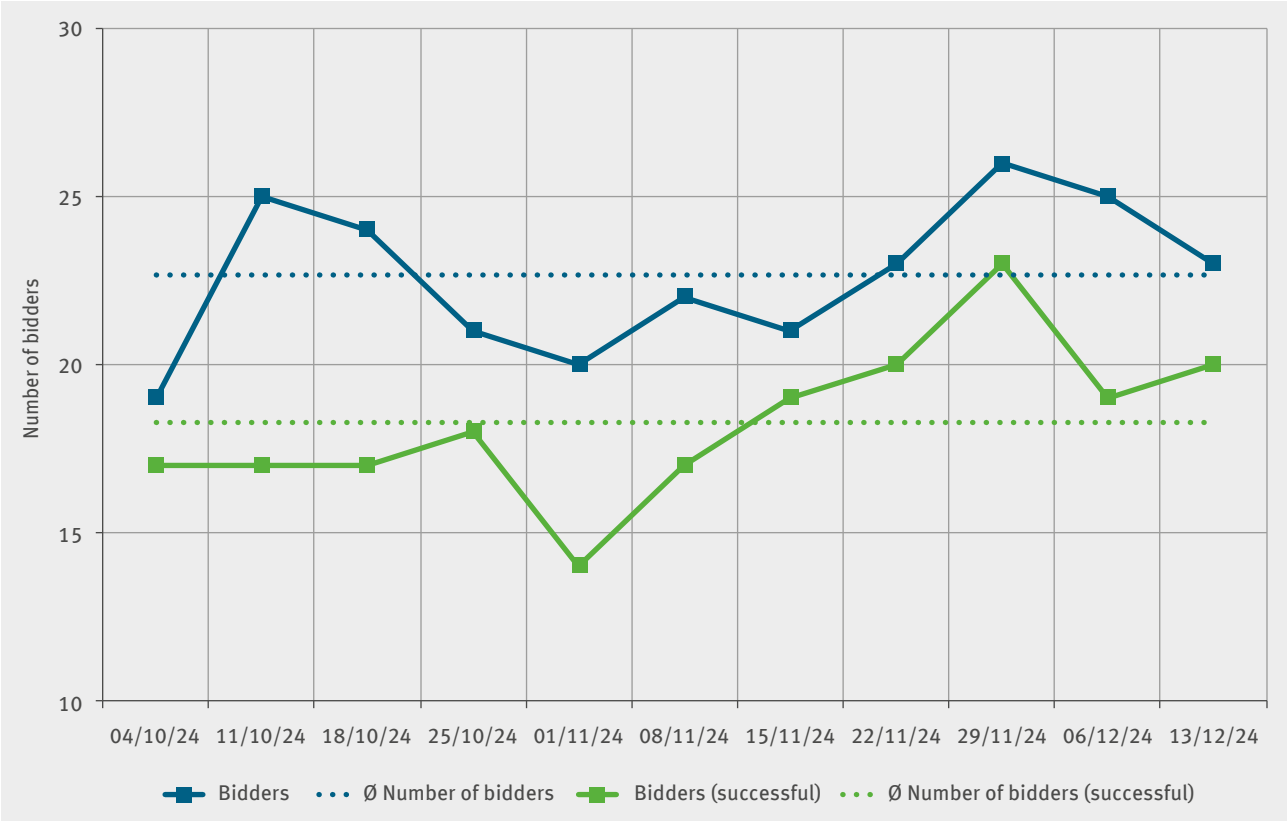
2 Bidder Participation

2.1 Participation

The number of participants in the German EUA auctions in the fourth quarter averaged across all dates was about 23 bidders. The average number of successful bidders was about 18. Participation in the auctions fluctuated between 19 and 26 bidders during the reporting period. The number of successful participants ranged from 14 to 23. There was no auction in which all participating bidders were successful.

With 15 bidders, the participation in the aviation allowances auction on 16th October was below the average participation in the EUA auctions.

The following figure illustrates EUA auction participation over time.



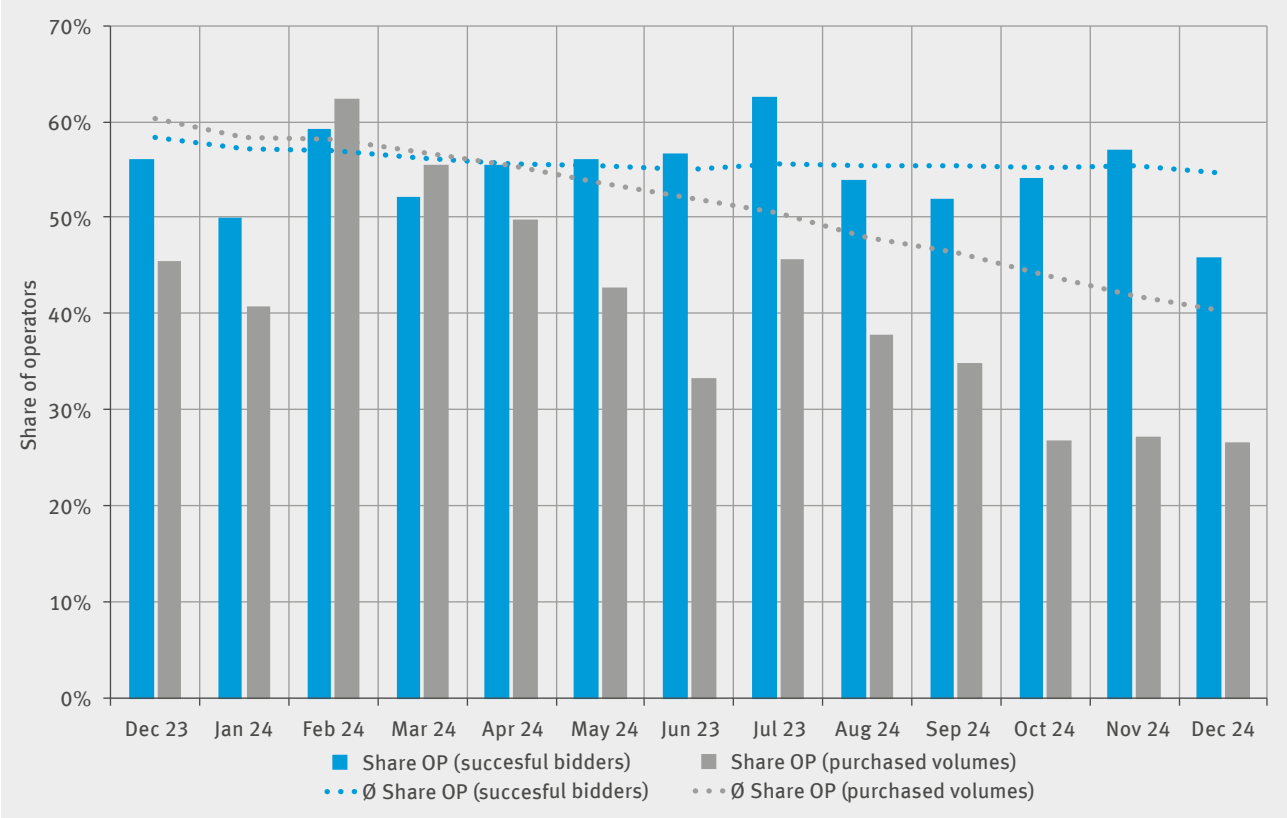
Source: EEX, DEHSt

Figure 2: Number of participating and successful bidders per auction date (EUA)

By the end of December 2024, a total of 111 trading members of EEX (plus 1 compared to the previous month) had been authorized to submit bids in the primary market auctions. In December, a total of 24 different bidders were successful in the German primary market auctions. Out of the 24 of those bidders, 11 were operators (i.e. operators of ETS covered installations or aircraft operators), which accounts for 46 percent of all successful bidders.[6] Overall, operators purchased approximately 27 percent of the auctioned volumes in December 2024. Thus, the share acquired by operators during the weekly auctions was below 50 percent for the ninth month in a row and at a similar level to the previous two months, its lowest since August 2016. The moving averages of the past 12 months for both parameters are at around 55 and 40 percent respectively. The average values of the volume shares show a clear downward trend.

6 Pursuant to Art. 18 of the EU Auctioning Regulation bidding categories eligible to take part in the auctions at EEX are amongst others (installation and aircraft) operators, credit institutions, investment firms and business groupings.

The following figure illustrates the development of the proportion of operators among the successful bidders as well as of the purchased auction volumes in the past 12 months before the current reporting period.



Source: EEX, DEHSt
OP = Operators

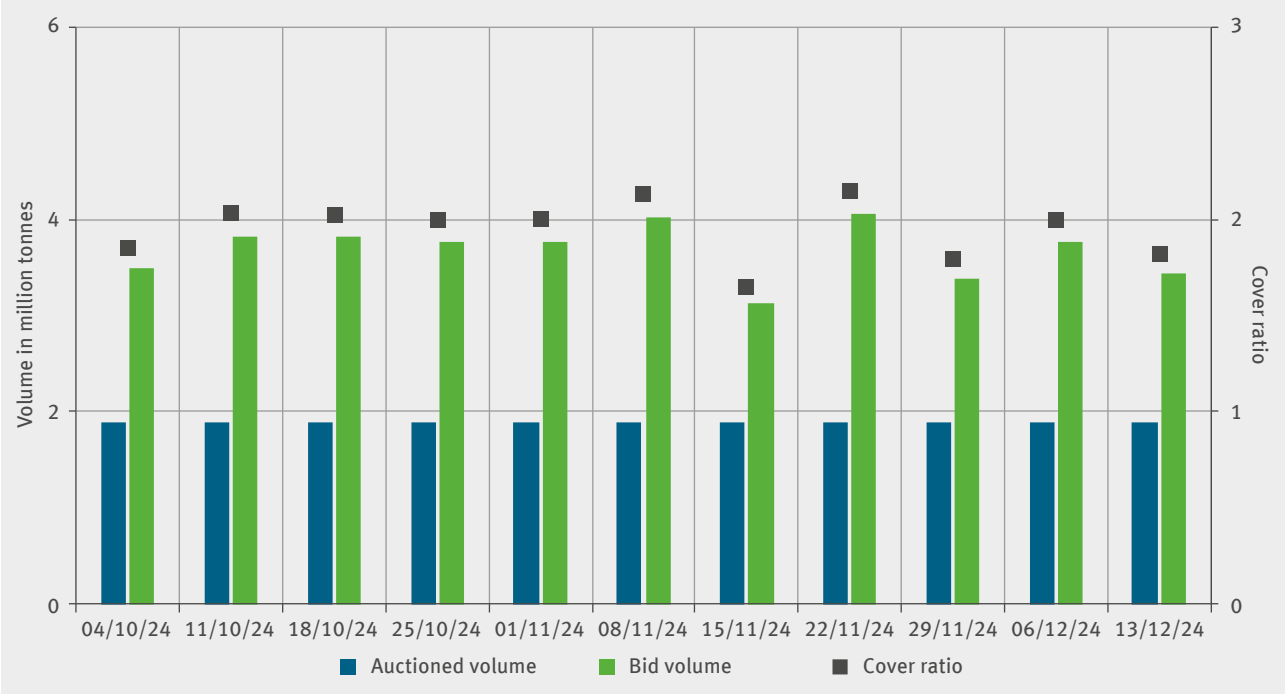
Figure 3: Share of operators among successful bidders and purchased auction volumes over the past 12 months (EUA)

2.2 Volume Demand

During the reporting period, the aggregated bid volumes per auction date fluctuated from 3.1 to 4.1 million allowances. Overall, valid bids totalling about 40 million allowances were entered into trading system in the fourth quarter. The supply volumes were stable at around 1.9 million allowances per date. The resulting cover ratios during the reporting period ranged between 1.65 and 2.15 (1.95 on average).

The cover ratio in the EUAA auction on 16th October was 3.03 and thus above the fluctuation range of the cover ratios in EUA auctions.

The figure below summarises the results for each EUA auction date.



Source: EEX, DEHSt

Figure 4: Auctioned and bid volume and cover ratio for each auction date (EUA)

Thus, the average bid volume per EUA auction participant in the fourth quarter was about 160,000 allowances. The average volume awarded to each successful bidder was about 105,000 allowances.

In the EUAA auction on 16th October the average bid volume per auction participant was about 215,000 allowances. The average volume awarded to each successful bidder was about 175,000 allowances.

Awarding allowances at all auctions is based on the EU-wide applicable uniform price procedure with one prescribed round of bidding (also see Info box below).

Uniform Price Auction at the EEX

The auctions at the EEX follow the uniform price procedure which is common practice throughout the European Union. It involves one round of bidding and a closed order book. This simple and robust procedure protects against market manipulation and was used already for German auctions during the second trading period. The uniform price procedure means that all successful bidders pay the same price. The auction clearing price is determined by ranking all eligible bids, beginning with the highest bid, according to the bidding price.

In the event of equal bids, the bids are sorted by a random process. The bid volumes are aggregated starting with the highest bid until the volume of emission allowances on offer (e. g. 1.9 million EUA) is reached. The bid price at which the aggregate volume of bids reaches or exceeds the offered volume of auctioned allowances determines the auction clearing price.

Bidders will thus be awarded their bid volume if their bidding price at east matches the auction clearing price. If a bid matches the clearing price exactly, the bidder may be awarded the remaining allowances which may be below the volume requested, depending on the bidding situation. If the bid from several participants matches the auction clearing price exactly, the above-mentioned random process applies.

3 Auctions and the Secondary Market

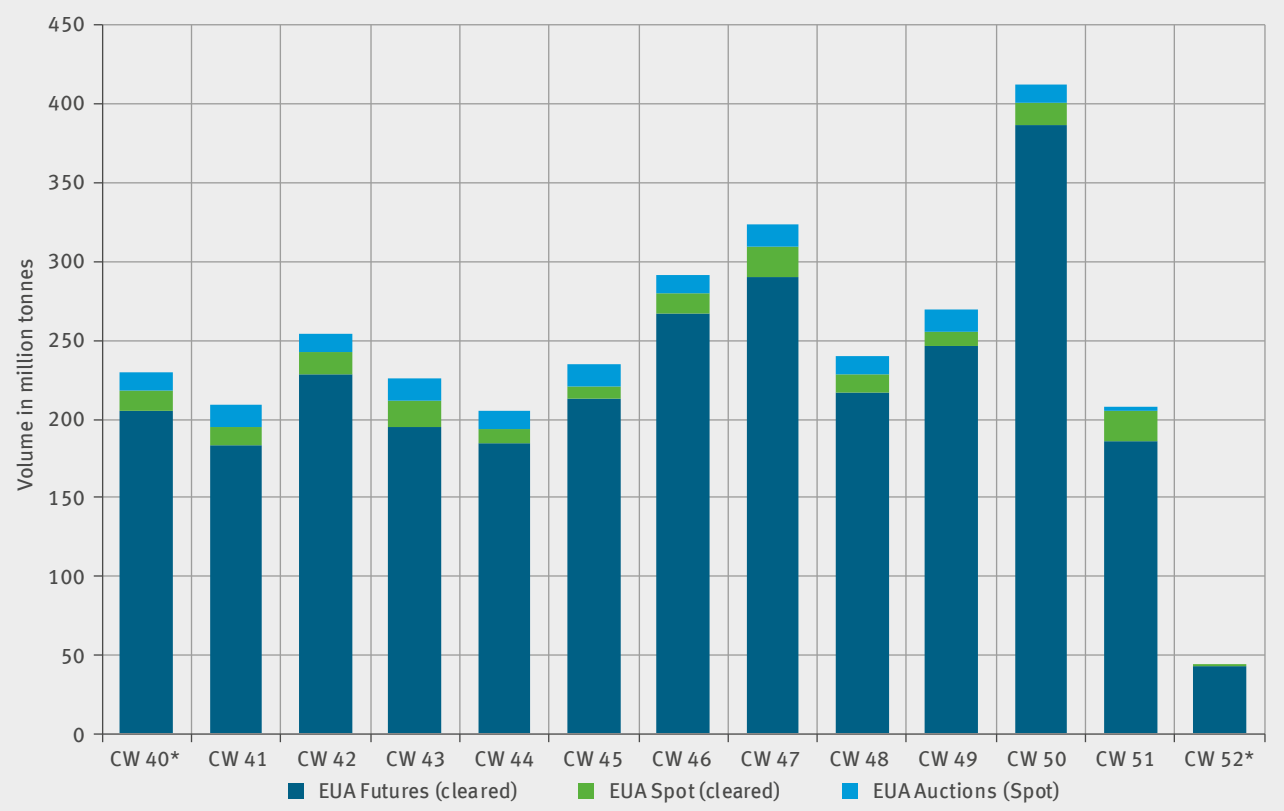
3.1 Trading Volume EUA

In the fourth quarter of 2024, a total of about 21 million EUA were auctioned at EEX for Germany. During the reporting period, the cumulative trading volume in the relevant total market was about 3.15 billion EUA.[7] This comprises the monthly data of the London Energy Brokers' Association (LEBA) (see also Figure 7).[8]

The secondary market trading volume of 3.01 billion EUA was divided into approximately 2.85 billion EUA on the cleared futures market, about 158 million EUA on the cleared spot market and 2 million EUA bilaterally settled through the LEBA-Brokers.

In the fourth quarter of 2024, a total of about 144 million allowances were auctioned by Member States on the primary market as spot contracts. With respect to the relevant total market for EUA, around 5 percent of the traded volume was related to auctions (see also Table 3).

The figure below illustrates the trading volumes separately for different market segments based on calendar weeks (CW).



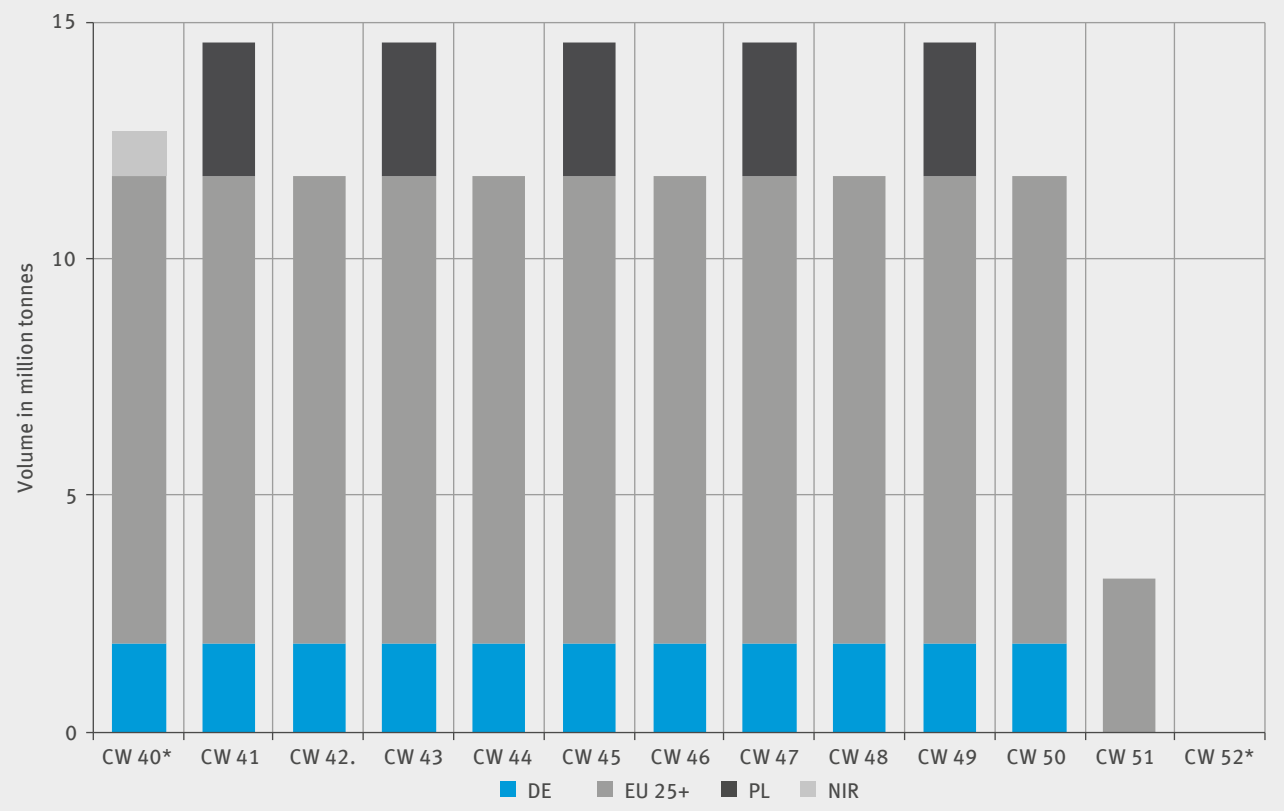
Source: ICE, EEX, Refinitiv Eikon, DEHSt
* CW 40: 30/09/2024–04/10/2024; CW 52: 23/12/2024–27/12/2024

Figure 5: Trading volumes on the primary and secondary market in the fourth quarter of 2024 (EUA)

7 All emission allowances (EUA) of the fourth trading period traded on the spot and futures markets on the ICE, EEX and Nasdaq OMX exchanges are considered. Transactions completed directly on the stock exchange (screen traded) or those bilateral transactions that at least used the stock exchange clearing mechanism are included. The EUA spot contracts auctioned in the fourth quarter of 2024 are also considered. As far as data on volumes from bilateral trading activity (i.e. off-exchange) is available, it is included in the analysis. Option contracts are not involved.

8 LEBA is an association of major brokerage firms involved in trading on the energy markets. Their monthly volume reports also include bilaterally settled contracts that were not registered on clearing platforms ("OMP Bilaterally Settled Volume"). The data are not broken down into futures-/spot-contracts. Moreover, since the volumes are only available as *monthly* aggregated values, the data were taken into account only in Figure 7, but not in Figure 5.

The following figure shows the auction volumes on the primary market per calendar week (CW). In addition to the German auctions, auction volumes of other EU Member States, as well as Norway, Iceland, Liechtenstein, Northern Ireland[9], for the EU Innovation/Modernisation funds and Recovery and Resilience Facility (RRF) are also included in the figure.

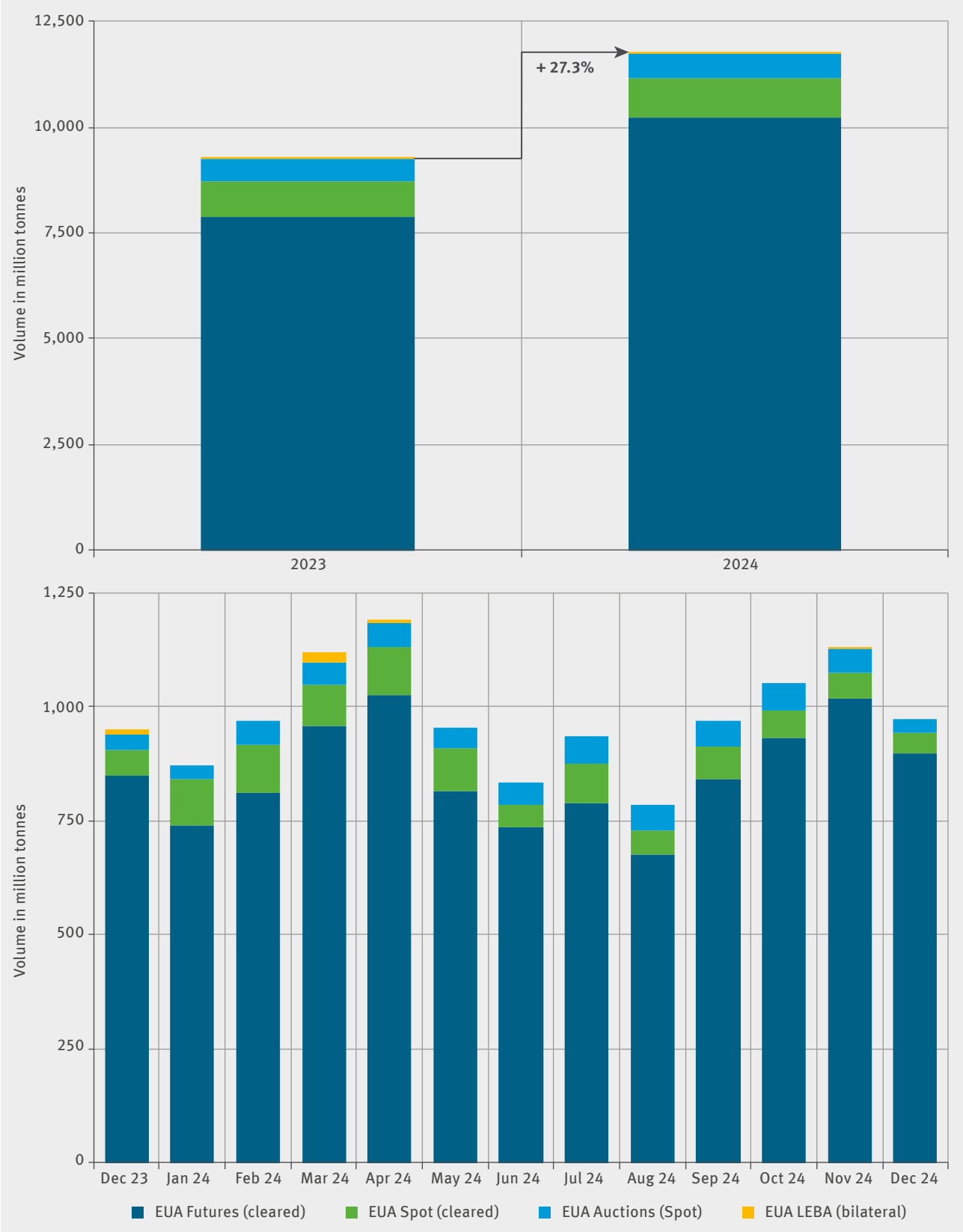


Source: EEX, Refinitiv Eikon, DEHSt
DE = Germany, PL = Poland, EU25+ = EU-Commission on behalf of other member states and Norway, Iceland and Liechtenstein, NIR = Northern Ireland; EU Innovation/Modernisation funds, Recovery and Resilience Facility (RRF)
* CW 40: 30/09/2024–04/10/2024; CW 52: 23/12/2024–27/12/2024

Figure 6: Auction volumes on the primary market in the fourth quarter of 2024 (EUA)

9 In 2021, the UK left the EU. However, pursuant to the Protocol of Ireland and Northern Ireland, the EU ETS 1 continues to apply to generation of electricity in Northern Ireland. Therefore, a small volume of allowances will be auctioned on the EEX on behalf of the UK, corresponding to its share in the Union-wide cap.

The following chart shows the development of the monthly EUA trading volumes on the primary and secondary market over the past 12 months before the current reporting month. In 2024, the total market volume of about 11.79 billion EUA was about 27 percent above the previous year's level (2023: 9.26 billion EUA).



Source: ICE, EEX, Nasdaq OMX, Refinitiv Eikon, LEBA, DEHSt

Figure 7: Trading volumes on the primary and secondary market over the past 12 month (EUA)

Table 3: Market shares of the various trading segments (EUA)

Market segments		Dec 2024	Nov 2024	2024	2023
Primary market	EUA Auctions (Spot)	3.0%	4.8%	5.0%	5.6%
Secondary market	EUA Futures (cleared)	92.3%	90.1%	86.9%	85.2%
	EUA Spot (cleared)	4.6%	4.9%	7.8%	9.1%
	EUA LEBA (bilateral)	–	0.2%	0.3%	0.1%

Source: ICE, EEX, Nasdaq OMX, Refinitiv Eikon, LEBA, DEHSt

3.2 Trading Volume EUAA

In the fourth quarter of 2024, no aviation allowances (EUAA) were traded on the secondary market[10]. During the same period, about 2.2 million EUAA were auctioned by the countries participating in the EU ETS 1 on the primary market. Further information on the EUAA auction calendars is available on the website of the European Commission.

3.3 Auction Prices and Price Development on the Secondary Market

Reference prices from the most liquid trading venues served as appropriate benchmarks for evaluating the auction clearing prices achieved on the EUA spot auctions. The reference contract for the auctions is the daily future traded on the ICE Endex or the EEX EUA spot contract. The product specifications are comparable with the spot-product of the auctions. The trade price immediately before the end of the bidding period was used as a reference price. If no trades take place immediately before the end of the bidding period, the last best bid was used as a reference price (see also Info box below).

Reference Prices

In regular exchange trading (secondary trading), participants enter their buying and selling requests (bids and asks) into an order book which is constantly updated. As in auctions, such buying bids in secondary trading include quantity and price limits. The price limit is the highest price a buyer is prepared to pay for the purchase of an allowance. Sellers can also put a price limit on their bid, i. e. a minimum price per allowance sold.

The order book is open to inspection by any participant. It shows the current highest purchase offer (best bid) and the lowest sales offer (best ask) with the relevant quantities. Trading occurs where best bid and best ask overlap. Trading price records finally show the price limits where supply equals demand.

10 All aviation allowances (EUAA) traded on the spot and futures market on EEX are considered. Transactions completed directly on the stock exchange (screen traded) or those bilateral transactions that at least used the stock exchange clearing mechanism are included. By the end of 2022, ICE has ceased trading EUAA contracts.

Table 4 shows the relative deviations of the auction clearing price from the secondary market. The deviation on each EUA auction date just before the end of the bidding period (11:00 am) varied over the reporting period between € –0.24 (–0.36 percent) and € +0.01 (+0.01 percent).

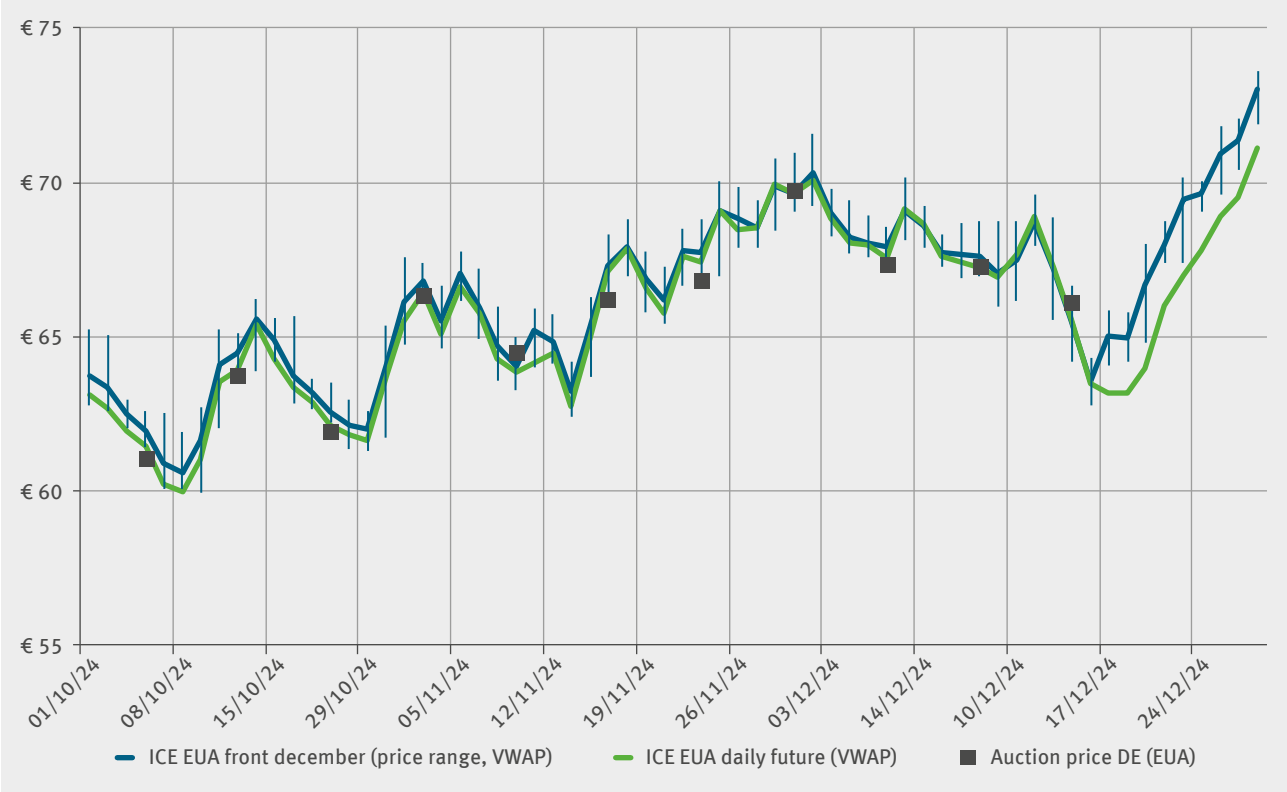
For a mean value analysis, the arithmetic mean of the absolute values of individual deviations over the reporting period is calculated. The average deviation from the last best bid was € 0.11 or 0.17 percent averaged over all dates.

Table 4: Auctions and deviations from secondary market prices the fourth quarter of 2024 (EUA)

Date	Contract	Auction price	Deviation from secondary market*	
			Absolute	%
04/10/2024	EUA Spot 4th TP	€ 61.03	€ –0.14	–0.23%
11/10/2024	EUA Spot 4th TP	€ 63.71	€ –0.09	–0.14%
18/10/2024	EUA Spot 4th TP	€ 61.90	€ –0.07	–0.11%
25/10/2024	EUA Spot 4th TP	€ 66.32	€ –0.08	–0.12%
01/11/2024	EUA Spot 4th TP	€ 64.46	€ –0.08	–0.12%
08/11/2024	EUA Spot 4th TP	€ 66.20	€ –0.07	–0.11%
15/11/2024	EUA Spot 4th TP	€ 66.78	€ –0.24	–0.36%
22/11/2024	EUA Spot 4th TP	€ 69.71	€ –0.08	–0.11%
29/11/2024	EUA Spot 4th TP	€ 67.30	€ –0.16	–0.24%
06/12/2024	EUA Spot 4th TP	€ 67.25	€ 0.01	0.01%
13/12/2024	EUA Spot 4th TP	€ 66.07	€ –0.24	–0.36%
Mean absolute deviation			**€ 0.11**	**0.17%**

Source: ICE, EEX, Refinitiv Eikon, DEHSt
* best bid ICE Daily Future

The following figure shows an overview of the price development for emission allowances on the secondary market. In addition to the daily futures the development of futures for December delivery of the current and the following year is also shown (so-called “front-December future”).[11] Rather than following the common practice of considering only settlement prices, our analysis takes into account all trades completed at the exchanges.[12] The graphic is based on volume-weighted daily average trading prices (VWAP).[13] In addition, the daily trading range of the front-December contract is also shown in the chart (vertical price line). In the fourth quarter of 2024 the price of the front-December contract fluctuated within an interval between € 59.95 and € 73.62.



Source: ICE, EEX, Refinitiv Eikon, DEHSt
DE = Germany

Figure 8: ICE secondary market prices and EEX auction clearing prices (EUA)

11 ICE EUA Dec24 contract expired on 16/12/2024.

12 Settlement prices are determined by the exchanges at the end of the trading day. Hence, they only reflect trade activity immediately before closure of the exchange and are, thus, not always an adequate reflection of the prevailing intra-day prices in the secondary market.

13 The volume-weighted average prices (VWAP) for the ICE EUA front-December future and the ICE daily future contract is based on data from Refinitiv Eikon.

4 Implementation of Valid Regulations

4.1 Surveillance Report by HÜSt

In order to guarantee above-board market activities are free of manipulation on the EEX continuous surveillance is conducted by an independent Market Surveillance Office (Handelsüberwachungsstelle, HÜSt) as required under German law. HÜSt reports to the DEHSt on the results of its surveillance activities through regular internal reports. HÜSt reported on auctions in the fourth quarter of 2024 as follows (excerpt):

"Within their surveillance for the auctioning of emission allowances on behalf of the German Environment Agency at the EEX in the fourth quarter of 2024 the Market Surveillance Office of the EEX could not detect any irregularities and especially no behaviour of bidders that was targeted at manipulating the auction price."

4.2 Fair and Open Access

The implementation of the provisions of the EU Auctioning Regulation guarantees a fair and open access for all auction participants at EEX. Current access options can be found at EEX Website (Access to the auctions).

More information about the auction process is also available on the EEX auction hotline. EEX can be reached by phone from Mondays to Fridays from 8:00 to 18:00 CET at +49 341 2156-331 or by e-mail at auctions@eex.com.

4.3 Auction Result Publication

All auction results and accompanying data were made publicly available during the reporting period within a 15-minute time window directly linked to the respective auction at EEX Website.

4.4 Other Events

There are no matters relating to the implementation of the existing rules to be reported during the reporting period. The current regulations can be downloaded from EEX Website (Rules and Regulations).

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